

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q21-1962

Vc Porterdale Hydroelectric 138 kV
November 03, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1962. The Interconnection Customer proposes a 141.1 MW solar photovoltaic facility interconnecting at the Vc Porterdale Hydroelectric 138 kV substation in the Virginia area. The request was received on December 24, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 506 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$241,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
ROANOKE ELECTRIC STEEL / MUDDY RANCH SUBSTATION 138.0 kV L	HUNTINGTON CURT 345 kV/138 kV XFMR	120.4	338	50.6
STEABOAT / MUDDY RANCH 138.0 kV	HUNTINGTON COURT 345/138 kV TX	102.7	357	28.1

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
ROANOKE ELECTRIC STEEL / MUDDY RANCH SUBSTATION 138.0 kV L	Rebuild line with higher-capacity conductor	\$1,328,000	11% (\$143,000)
STEABOAT / MUDDY RANCH 138.0 kV	Reconductor line	\$1,207,000	8% (\$98,000)
	Total allocated to project		\$241,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.